

RISK RATING 2.0 • APRIL 2023

Safer Ground, Tighter Markets

The public was paying for the water. Subsidized insurance made floodplain houses cheap, and the fiscal bill stayed with the government.

Does the shock stop at the floodplain, or does it tighten safer ground too?

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02 • WHAT THE OLD PREMIUM LEFT BEHIND

A private loss, and a public one.



03 • THE INSURER

+82%
premium component
Allegheny County

~18%
annual cap. Buyers
capitalize the full path.

04 • INSIDE THE FLOODPLAIN

The price falls first.
The houses leave later.

-28.9% -33.5%

price on impact \$110,000 → \$78,242
long-run stock 4,036 → 2,686
Long-run price \$78,340 (-28.8%). The stock cannot jump, so the price does. $\psi_{H,t}$ does not change.

05 • SAFER GROUND

No new premium.
The county still tightens.

+0.15%

Long-run price, \$218,000 → \$218,326. Stock 379,126 → 379,877.
On impact, \$218,117 (+0.054%). County stock 383,162 → 382,493 (-0.17%). Safer ground is already in place in Allegheny.

06 • THE MODEL

One rent. Two building sites.

$$R(S) = \frac{\alpha(W - \beta_0)}{S - \beta_1(I - \alpha)} \quad S = S_{NF} + S_{FP}$$

$$Y_t(P_t) = a_t P_t^f$$

$$\dot{P}_{NF} = \psi_{NF} P_{NF} - R(S) \quad \dot{P}_{FP} = \psi_{FP}(m) P_{FP} - R(S)$$

$$\dot{S}_{NF} = a_{NF} P_{NF}^f - \delta S_{NF} \quad \dot{S}_{FP} = a_{FP} P_{FP}^f - \delta S_{FP}$$

$\psi_{FP}(m) = \psi_{FP}^0 + f m$. The shock is an 82% rise in the premium component, not $m = 0.82$.
 $\psi_{NF} = 0.059$, $\psi_{FP}^0 = 0.1169$, $f = 0.0579$, $\epsilon = 0.08$, then $1.20 \cdot 5 = 0.02$.
If $\alpha = 0.5$, the safer long-run price rises 9.9%, not 0.15%.

07 • MIAMI-DADE • 85% PREMIUM INCREASE

Half the homes sit in the flood.

+2.60%
X outside
stock +3.13%

-5.25%
A inland
stock -6.20%

Long-run prices. About 50.5% of homes are in A or V. Short run: X +1.04%, A -6.79%, V -17.94%.

-16.68%
V coastal
stock -19.67%

08 • POLICY

Price the risk. Make room on higher ground.

Keep property-level premiums. Pair them with supply where building is safe, or the region ends with fewer houses.
Counterfactuals, not causal estimates. Losses are not yet quantified. Next: every U.S. county, and α .